

Name: Ram Subhash

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PROFESSIONAL SUMMARY:

- Processed and transformed large-scale data using **Snowflake**, **SnowSQL**, and **Snowpipe**, implementing ETL workflows through **dbt**, **Matillion**, and **Talend** to ensure clean, consistent data pipelines feeding **Data Marts** and **Snowflake schema** for analytics and reporting across multiple cloud and on-premise environments.
- Executed batch and streaming pipelines using **Python**, **PySpark**, **Pandas**, **NumPy**, and **Apache Spark**, integrating real-time ingestion from **Apache Kafka** and **Apache Pulsar** to deliver high-quality, timely operational and business datasets to downstream teams and visualization tools.
- Orchestrated workflow scheduling and pipeline automation using **Airflow**, **Luigi**, and **Prefect**, coordinating incremental updates through **Snowflake Tasks** and **Snowflake Streams** while maintaining job reliability, retry mechanisms, and monitoring to reduce processing latency across analytics pipelines.
- Built cloud-based data integration using **AWS S3**, **AWS Redshift**, **AWS Glue**, **AWS Lambda**, **AWS EC2**, **Azure Data Factory**, **Azure Synapse**, **Azure Blob Storage**, **GCP Cloud Storage**, **GCP BigQuery**, and **GCP Dataflow**, ensuring seamless ingestion, transformation, and storage of structured and semi-structured logistics and business data.
- Designed interactive dashboards and data visualizations using **React.js**, **Redux**, **Zustand**, **Plotly**, **Chart.js**, **D3.js**, and **Axios**, delivering operational, financial, and predictive insights while connecting with backend APIs and **FastAPI** or **Flask** services.
- Managed data quality, lineage, and governance using **Collibra**, **Alation**, **Apache Atlas**, **Erwin**, and **Power Designer**, documenting **Dimensional Modeling**, **Star schema**, and **Snowflake schema** structures for audit compliance and ensuring accurate analytics across multiple datasets and **Data Marts**.
- Monitored pipelines and infrastructure performance using **Datadog**, **CloudWatch**, **New Relic**, and **Azure Monitor**, implementing alerting and reporting mechanisms to track latency, job failures, and resource usage for operational reliability in cloud and on-prem environments.
- Developed, cleaned, and transformed datasets using **Python**, **PySpark**, **Pandas**, and **NumPy**, preparing training and operational data for **scikit-learn**, **TensorFlow**, **PyTorch**, and **MLflow**, while enabling robust machine learning experimentation and predictive modeling.
- Automated code integration, deployment, and testing using **Git**, **GitHub**, **GitHub Actions**, **GitLab**, **GitLab CI/CD**, and **Jenkins**, maintaining version-controlled **dbt** models, ETL scripts, and API workflows for repeatable and consistent deployments across environments.
- Optimized **SQL** and **SnowSQL** queries, performing complex joins, aggregations, and window functions to generate analytics-ready datasets, improving processing efficiency and reducing latency for real-time reporting and business intelligence dashboards.
- Designed reusable APIs using **FastAPI**, **Flask**, and **Fetch API**, integrating processed datasets into interactive dashboards and analytics tools to provide end-users with up-to-date metrics and actionable insights in logistics, finance, and operations teams.
- Coordinated project and team collaboration using **Jira**, **Trello**, **Slack**, **Zoom**, and **Confluence**, managing task assignments, documentation, and cross-functional communications to ensure timely delivery of ETL workflows, models, and analytics outputs for operational and strategic decisions.
- Maintained and provisioned infrastructure using **Terraform** and **CloudFormation**, deploying pipelines, cloud storage, and compute resources efficiently while following reproducible, version-controlled practices to support scalable data operations and analytics platforms.

- Implemented **Snowflake Cortex AI** to enrich datasets with intelligent insights and predictive outputs, integrating with **Snowflake schema**, **Data Marts**, and **dbt** models to enhance analytics capabilities and decision-making for operations and business teams.
- Standardized **Dimensional Modeling** and **Star schema** designs for multiple datasets, creating structured **Data Marts** for logistics, customer, and operational data to ensure consistency and support advanced analytics, reporting, and predictive modeling initiatives.
- Collaborated across cloud platforms including **AWS**, **Azure**, and **GCP**, combining tools such as **AWS Redshift**, **Azure Synapse**, and **GCP BigQuery** to deliver harmonized data pipelines, ensuring secure, reliable, and timely availability of enterprise datasets.
- Integrated real-time and batch workflows using **Apache Kafka**, **Apache Pulsar**, **Snowflake Tasks**, **Snowflake Streams**, and **dbt**, enabling operational teams to access updated metrics, KPIs, and predictive analytics for business decisions in logistics, finance, and operations.
- Designed machine learning experiments using **scikit-learn**, **TensorFlow**, **PyTorch**, and **MLflow**, processing cleaned datasets from **Python**, **PySpark**, and **Snowflake**, enabling predictive models and AI-driven insights for operational optimization and customer analytics.

TECHNICAL SKILLS:

- **ETL:** Informatica Power Center 10.x/9.6, AWS Glue, Talend 5.6, SSIS, Semarchy xDM
- **Databases & Tools:** MS SQL Server 2014/2012, Teradata 15/14, Oracle 11g/10g, Erwin 8/9, MySQL
- **Cloud Environment:** AWS (Snowflake, RDS, Redshift, EC2, EMR, S3, Lambda, Glue, Athena, SQS, VPC, Cloud Watch, Jenkins), Azure (HDInsight, CosmosDB, Fabric, Blob storage, Data lake, Data factory, SQL, Databricks), GCP (DataProc, Bigquery, Cloud storage, GKE, Functions, Spanner)
- **Reporting Tools:** Tableau, PowerBI, EHR, Semarchy
- **Big Data Ecosystem:** HDFS, Hive, MapReduce, Sqoop, Flume, Spark, Spark Streaming, NiFi, Yarn, Zookeeper, Airflow, Kafka
- **Programming languages:** Unix, SQL, Python, T-SQL, Java, Pyspark, Spring boot, Scala
- **Software Life Cycle/ Methodologies:** SDLC, Waterfall, and Agile models
- **Containerization and Orchestration Tools:** Kubernetes, Docker, Jenkins, Docker Hub
- **Python Libraries / Machine Learning Algorithms:** Pandas, NumPy, Matplotlib, Scikit-Learn, Decision Tree, Random Forest, Logistic Regression, Gradient Boosting
- **Data Warehousing & BI:** Star and Snowflake schema, Facts and Dimensions tables, SSIS, Splunk, Cloudera

PROFESSIONAL EXPERIENCE:

Client: State Farm

Role: SR. BIG DATA ENGINEER

Responsibilities:

Location: Bloomington, IL

October 2022 – Current

- Designed and implemented end-to-end ETL pipelines using **Python**, **PySpark**, and **Apache Spark**, ingesting large-scale data from **AWS S3** into **Snowflake**. Optimized transformations with **Pandas** and **NumPy** to enable efficient data processing and reporting.
- Managed real-time data streaming workflows by integrating **Apache Kafka** and **Apache Pulsar** with **Snowflake Streams** and **Snowpipe**, ensuring continuous data ingestion into **Snowflake** and supporting timely analytics for business-critical dashboards.

- Developed and deployed automated ETL processes using **dbt** and **Matillion**, performing transformations within **Snowflake**, while maintaining consistent schema structures such as **Star schema** and **Data Marts** for analytical consumption.
- Tuned **Snowflake** performance by optimizing clustering keys, partitioning strategies, and caching, while using **SnowSQL** to execute high-performance queries and monitor workloads, improving query response time and reducing compute costs.
- Built machine learning pipelines with **scikit-learn**, **TensorFlow**, and **PyTorch**, orchestrating experiments via **MLflow**, and deployed predictive models integrated with **Snowflake Cortex AI** for enterprise-wide AI-enabled insights.
- Automated workflow orchestration using **Airflow** and **Jenkins**, scheduling ETL/ELT jobs from **AWS Glue** and **AWS Lambda**, ensuring fault-tolerant execution and maintaining data integrity across cloud environments.
- Designed and implemented secure data architectures using **Terraform**, **AWS IAM**, and **Snowflake role-based access control**, while monitoring system performance with **Datadog** and **CloudWatch** to ensure compliance and operational stability.
- Modeled enterprise data structures through **Dimensional Modeling** using **Erwin**, defining **Star schema** and **Data Marts**, enabling seamless integration with **Snowflake** for analytics and BI reporting.
- Developed interactive dashboards using **React.js**, **Redux**, **Zustand**, and **Plotly**, fetching data through **Axios** APIs from backend services deployed on **Flask**, providing users with responsive, data-driven front-end applications.
- Implemented CI/CD pipelines for data engineering workflows using **Git** and **Jenkins**, automating version control, code validation, and deployment of **dbt** models and **Python** scripts across cloud environments.
- Orchestrated cloud-based data ingestion from **AWS S3** to **AWS Redshift** and **Snowflake** using **AWS Glue**, performing schema transformations and data quality checks to maintain high-quality enterprise datasets.
- Monitored and optimized pipeline performance by leveraging **Datadog** and **CloudWatch** to track metrics, detect anomalies, and tune **Apache Spark** jobs and **Snowflake** queries for improved throughput and reliability.
- Integrated AI/ML predictions into enterprise workflows via **Snowflake Cortex AI**, **Python**, and **MLflow**, ensuring that processed datasets in **Snowflake** were ready for downstream analytics and reporting in dashboards.
- Developed and maintained complex **PySpark** and **Python** scripts for batch and streaming ETL jobs, combining **Pandas** and **NumPy** for preprocessing, and loaded processed datasets into **Snowflake** for analytical use.
- Collaborated with cross-functional teams using **Jira** and **Slack** to define requirements, design pipelines, and deliver production-ready **Snowflake** solutions, ensuring alignment with business objectives and compliance policies.
- Implemented data governance and lineage tracking using **Collibra** and **Informatica**, maintaining metadata, validating transformations, and enforcing enterprise-wide standards for security, compliance, and quality.
- Created reusable ETL templates in **dbt** and **Matillion**, standardized **Data Marts**, and applied **Dimensional Modeling** principles to support scalable analytics and reduce development cycle times.
- Orchestrated event-driven processing using **Snowflake Tasks**, **Snowflake Streams**, and **Apache Kafka**, ensuring real-time updates were reflected across **AWS** environments, while maintaining fault tolerance and end-to-end data reliability.

Environment: Python, PySpark, Apache Spark, AWS S3, Snowflake, Pandas, NumPy, Apache Kafka, Apache Pulsar, Snowflake Streams, Snowpipe, dbt, Matillion, Star schema, Data Marts, SnowSQL, scikit-learn, TensorFlow, PyTorch, MLflow, Snowflake Cortex AI, Airflow, Jenkins, AWS Glue, AWS Lambda, Terraform, AWS IAM, Datadog, CloudWatch, Dimensional Modeling, Erwin, React.js, Redux, Zustand, Plotly, Axios, Flask, Git, AWS Redshift, Jira, Slack, Collibra, Informatica, Snowflake Tasks

Client: Delta Airlines
Role: SR DATA ENGINEER
Responsibilities:

Location: Atlanta, GA
December 2019 – September 2022

- Designed and implemented scalable ETL pipelines using **Snowflake**, **Snowpipe**, **SnowSQL**, and **Fivetran**, automating real-time ingestion of flight and operational data. Optimized loading processes for minimal latency while ensuring data integrity and consistency across multiple **Snowflake schemas**.
- Developed modular transformation workflows with **dbt**, leveraging **Snowflake Tasks** and **Streams** to orchestrate incremental data updates. Implemented version-controlled transformations and integrated with **GitHub** and **GitHub Actions** for CI/CD deployment of production-ready pipelines.
- Engineered high-performance batch and streaming data pipelines using **Python**, **PySpark**, **Pandas**, **NumPy**, and **Apache Spark**, enabling large-scale processing of flight telemetry and booking data while maintaining efficient memory and compute resource utilization.
- Built real-time event processing solutions using **Apache Kafka** and **Apache Pulsar**, ensuring reliable ingestion of aircraft telemetry, passenger updates, and operational logs. Coordinated with **Prefect** to schedule, monitor, and retry failed pipelines automatically.
- Designed and maintained data integration workflows with **Azure Data Factory**, orchestrating data movement from **Azure Blob Storage** to **Azure Synapse** and **Snowflake**, ensuring near real-time availability for analytics and reporting across multiple aviation business units.
- Implemented robust monitoring and observability using **New Relic** and **Azure Monitor**, tracking pipeline health, latency, and error rates. Automated alerts for failed ETL jobs and performance degradation, enabling proactive resolution of data workflow issues.
- Developed dimensional models and data marts using **Dimensional Modeling** principles and **Snowflake schema** structures, providing consolidated flight operations, ticketing, and customer analytics datasets for downstream reporting tools.
- Designed RESTful APIs using **FastAPI** to serve processed flight and operational datasets to internal applications. Integrated API endpoints with **React.js**, **Redux**, and **Zustand** front-end dashboards, enabling interactive visualizations and real-time metrics.
- Developed dynamic dashboards and interactive charts using **Chart.js** and **Axios**, consuming processed datasets from **Snowflake** and APIs, enabling operations teams to track airline performance, flight delays, and passenger load factors efficiently.
- Performed advanced SQL development and query optimization using **SQL** and **SnowSQL**, creating complex joins, aggregations, and window functions to deliver business-ready datasets for reporting and analytics.
- Leveraged **ER/Studio** and **Alation** to document data lineage, metadata, and relationships across multiple **Snowflake** databases, ensuring compliance with governance standards and enabling accurate audit trails for aviation data.
- Automated CI/CD pipelines for ETL, dbt models, and API deployments using **GitHub Actions**, integrating testing, linting, and version control practices. Ensured seamless promotion of code from development to production environments.
- Developed reusable **Python** and **PySpark** libraries for data cleansing, enrichment, and transformation, standardizing workflows across flight operations, customer service, and cargo datasets, improving consistency and maintainability.
- Collaborated with cross-functional teams on real-time data strategies, implementing **Prefect** orchestrations for scheduled and event-driven tasks. Ensured timely data delivery to analytics platforms and operational dashboards.

- Utilized **Terraform** to provision cloud infrastructure, including **Azure Synapse** and **Azure Blob Storage**, following infrastructure-as-code principles. Automated environment setups and maintained reproducible, version-controlled deployments.
- Optimized streaming and batch pipelines to handle airline operational spikes using **Apache Spark**, **Kafka**, and **Pulsar**, reducing processing latency and improving system reliability during peak travel periods.
- Developed and maintained **dbt** models for multiple **Data Marts**, integrating historical and streaming data to support predictive analytics for flight schedules, revenue management, and passenger behavior analysis.
- Engaged in collaborative documentation and knowledge sharing via **Confluence** and **Teams**, maintaining process guides, workflow diagrams, and ETL runbooks to enable operational continuity and cross-team alignment in data engineering projects.

Environment: Snowflake, Snowpipe, SnowSQL, Fivetran, dbt, Snowflake Tasks, Snowflake Streams, GitHub, GitHub Actions, Python, PySpark, Pandas, NumPy, Apache Spark, Apache Kafka, Apache Pulsar, Prefect, Azure Data Factory, Azure Blob Storage, Azure Synapse, New Relic, Azure Monitor, Dimensional Modeling, FastAPI, React.js, Redux, Zustand, Chart.js, Axios, SQL, ER/Studio, Alation, Terraform, Confluence, Teams, Data Marts, Snowflake schema

Client: UPS

Location: Atlanta, GA

Role: DATA ENGINEER / DATA ANALYST

February 2017 – November 2019

Responsibilities:

- Designed and implemented scalable ETL pipelines using **Snowflake**, **SnowSQL**, **Snowpipe**, and **Talend**, automating ingestion of shipment and logistics data. Optimized loading processes for minimal latency while ensuring data consistency across multiple **Snowflake schemas** and **Data Marts**.
- Developed modular data transformation workflows with **dbt**, leveraging **Snowflake Tasks** and **Streams** for incremental updates. Integrated version-controlled transformations using **GitLab** and automated deployment via **GitLab CI/CD** for seamless production rollout.
- Engineered batch and streaming data pipelines using **Python**, **PySpark**, **Pandas**, **NumPy**, and **Apache Spark**, enabling large-scale processing of operational and shipment tracking data while maintaining optimal memory and compute efficiency.
- Built real-time event-driven data workflows with **Apache Kafka**, **Apache Pulsar**, and **Luigi**, ensuring reliable ingestion of logistics events. Scheduled and monitored pipelines for failures and retries, improving data freshness for reporting teams.
- Orchestrated cloud-based data integration using **GCP Dataflow** and **GCP Cloud Storage**, transferring processed datasets to **GCP BigQuery** for analytics. Automated workflows ensured near real-time availability of operational insights across logistics business units.
- Implemented monitoring and observability for pipelines using **Dynatrace** and **Stackdriver**, tracking job health, latency, and error rates. Automated alerts reduced downtime and improved incident response efficiency for ETL workflows.
- Designed dimensional models and **Data Marts** using **Dimensional Modeling** principles and **Snowflake schema**, consolidating shipping, inventory, and delivery data for downstream analytics and reporting dashboards.
- Developed RESTful APIs with **Fetch API** and backend processing in **Python**, serving operational and tracking data to internal dashboards. Integrated front-end visualizations using **React.js**, **Redux**, **Zustand**, and **D3.js** for interactive reporting.
- Built dynamic visual dashboards using **D3.js** and **Fetch API**, consuming transformed datasets from **Snowflake** and **GCP BigQuery**, enabling operations teams to monitor delivery performance and shipment delays in real time.

- Performed advanced data modeling and query optimization using **SQL** and **SnowSQL**, creating complex joins, aggregations, and window functions to deliver analytics-ready datasets for logistics and inventory teams.
- Leveraged **Power Designer** and **Apache Atlas** to document data lineage, relationships, and metadata across **Snowflake** databases, ensuring governance compliance and maintaining accurate audit trails for logistics operations.
- Automated CI/CD pipelines for **dbt** models, ETL workflows, and API deployments using **GitLab CI/CD**, integrating testing, linting, and version control to ensure seamless promotion from development to production environments.
- Developed reusable **Python** and **PySpark** scripts for data cleansing, enrichment, and transformation across logistics, inventory, and shipment datasets, standardizing workflows and improving maintainability and accuracy.
- Collaborated with cross-functional teams using **Trello** and **Zoom**, coordinating real-time data strategy, scheduling, and task management to ensure timely delivery of analytics insights for operational decision-making.
- Managed cloud infrastructure provisioning and reproducibility using **Terraform**, automating deployments of **GCP Cloud Storage** buckets, **Dataflow** jobs, and **BigQuery** datasets to support robust, scalable analytics pipelines.

Environment: Snowflake, SnowSQL, Snowpipe, Talend, dbt, Snowflake Tasks, Snowflake Streams, GitLab, GitLab CI/CD, Python, PySpark, Pandas, NumPy, Apache Spark, Apache Kafka, Apache Pulsar, Luigi, GCP Dataflow, GCP Cloud Storage, GCP BigQuery, Dynatrace, Stackdriver, Dimensional Modeling, Snowflake schema, Data Marts, Fetch API, React.js, Redux, Zustand, D3.js, SQL, Power Designer, Apache Atlas, Terraform, Trello, Zoom

Client: ADP

Location: Roseland, NJ

Role: DATA ENGINEER / MIGRATION SPECIALIST

January 2016 – January 2017

Responsibilities:

- Developed comprehensive **Payroll Data Integration** platform using **Azure Cloud**, improving data flow between **time-tracking**, **benefits**, and **payroll processing** modules while streamlining employee information management across multiple HR systems.
- Created data ingestion pipelines using **Talend ETL** and **Azure Data Factory** to process employee data from various corporate clients, implementing validation rules that ensured payroll accuracy for thousands of employees.
- Executed data migration projects transferring employee records from **legacy payroll systems** to modern **cloud-based platforms**, preserving historical payroll information while enabling new functionality through comprehensive data mapping.
- Built complex workflows with **Apache Airflow** to coordinate **ETL processes** across distributed payroll systems, orchestrating time-sensitive operations including pre-payroll validation and post-processing **tax calculations**.
- Implemented **Azure SQL Database** for structured employee records and **Azure Data Lake Storage** for semi-structured HR documents, reducing processing response times by 40% through strategic indexing and query optimization.
- Established data security protocols meeting **SOX compliance** and regulatory requirements for sensitive payroll data, implementing encryption, access controls, and audit logging to protect **personally identifiable information**.
- Created **predictive models** using **Azure Databricks**, improving workforce planning accuracy through **machine learning** techniques applied to historical employment data for optimized resource allocation.
- Built interactive **Tableau dashboards** visualizing payroll metrics and labor cost analytics, providing actionable insights into compensation trends and identifying cost-saving opportunities across departments.
- Developed custom **APIs** using **.NET Core** for secure data exchange between payroll systems and third-party **benefits providers**, enabling real-time updates while maintaining strict data security protocols.
- Created automated **tax filing** and reporting systems with **data quality monitoring** frameworks, ensuring timely submission of payroll tax documents while validating calculations against latest regulatory requirements.

Environment: Azure Cloud, Talend ETL, Azure Data Factory (ADF), Azure Databricks, Azure Data Lake Storage, Tableau, Epic

Client: Inno Minds

Location: Pune, India

Role: ETL DEVELOPER

May 2014 – November 2015

Responsibilities:

- Developed scalable **ETL pipeline** using **Oracle Data Integrator (ODI)** and **SQL Server Integration Services (SSIS)**, integrating data from client's **legacy systems** and third-party applications while maintaining consistent processing times during peak retail seasons.
- Set up **SQL Server data warehouse** with partitioning strategies and incremental loading techniques for secure management of raw, staging, and transformed **retail transaction data**, reducing processing windows for nightly batch operations.
- Used **Microsoft SQL Server Integration Services** to extract data from **Oracle**, **DB2**, and **MySQL** databases, implementing **change data capture** mechanisms that minimized system impact on production databases during business hours.
- Created **SSIS packages** processing **XML** and **Excel** file formats from **point-of-sale systems**, incorporating error handling routines and logging frameworks that improved troubleshooting capabilities while reducing data processing time by 40%.
- Built detailed data transformations with **Oracle Data Integrator**, focusing on **product hierarchy normalization** and **customer data cleansing**, standardizing product classifications across multiple retail brands and eliminating duplicate customer records.
- Set up **Microsoft SQL Server Analysis Services** for **multidimensional data modeling** with optimized aggregation designs, improving query performance for common retail KPI calculations and reducing report generation time by 30%.
- Developed **stored procedures** and **SQL scripts** to move transformed data into the enterprise data warehouse, implementing efficient bulk loading techniques while maintaining referential integrity across fact and dimension tables.
- Created automated ETL job scheduling systems using **SQL Server Agent** and **Windows Task Scheduler**, implementing dependency management and notification systems that ensured proper execution sequence while reducing manual intervention by 65%.
- Tracked ETL job performance through custom **monitoring dashboards** with alert systems for processing failures, providing real-time visibility into processing bottlenecks and reducing data availability issues by 45%.
- Built **Microsoft Power BI** connection to the SQL Server data warehouse, creating interactive dashboards enabling real-time analysis of **store performance** and **inventory metrics** that provided actionable insights for store managers.

Environment: AWS, Informatica, Talend, AWS S3, AWS Glue, AWS Redshift, Tableau, AWS Lambda

EDUCATION:

Bachelor of Technology (BTech) | Computer Science | Amrita Vishwa Vidyapeetham | Kerala, India | August 2010 - March 2014.

CERTIFICATIONS :

- **Salesforce Certified Sales Cloud Consultant**
- **Marketo Certified Solutions Architect (MCSA)**
- **HubSpot RevOps Professional Certification**
- **Pragmatic Institute: Certified Product Marketing Executive (Level VI)**